U–Th–Pb zircon ages of some Keweenawan Supergroup rocks from the south shore of Lake Superior

Robert E. Zartman, Suzanne W. Nicholson, William F. Cannon, and G.B. Morey

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Abstract: New single-crystal zircon U–Th–Pb ages for plutonic and rhyolitic Keweenawan Supergroup rocks from the south shore of Lake Superior provide geochronological constraints on magmatic evolution associated with the 1.1 Ga Midcontinent rift. Analyses of a granophyric phase of the Mineral Lake intrusion and the Mellen granite, both parts of the Mellen Intrusive Complex, and a laterally extensive rhyolite from the top of the Kallander Creek Volcanics have weighted average $^{207}Pb/^{206}Pb$ ages of 1102.0 ± 2.8 Ma (N = 2), 1100.9 ± 1.4 Ma (N = 5), and 1098.8 ± 1.9 Ma (N = 4), respectively. Analyses of a pyroclastic rhyolite flow at the top of the Porcupine Volcanics result in variable $^{207}Pb/^{206}Pb$ ages that range from 1080 to 1137 Ma. This rhyolite exhibits a continuum between morphologically complex and simpler prismatic zircon crystals, the latter yielding concordant analyses having a weighted average $^{207}Pb/^{206}Pb$ age of 1093.6 ± 1.8 Ma (N = 2). Four prismatic zircons from an aphyric rhyolite of the Chengwatana Volcanics in the Ashland syncline form a linear array intersecting concordia at 1094.6 ± 2.1 Ma (MSWD = 1.3). Another presumed Chengwatana rhyolite recovered from drill core intersecting the Hudson–Afton horst in southeast Minnesota yielded only ~20 morphologically indistinguishable zircons. Six analyses give $^{207}Pb/^{206}Pb$ ages ranging from 1112 to 1136 Ma, including one analysis with a virtually concordant age of 1130 Ma. This age, however, is considerably older than that obtained for the Chengwatana Volcanics in the Ashland syncline or any other precisely dated rock from the Midcontinent rift.

Résumé : Des âges U–Th–Pb nouvellement déterminés sur des cristaux uniques de zircon, extraits de roches plutonique et rhyolitique du Supergroupe de Keweenawan exposées sur la côte sud du lac Supérieur, permettent de fixer des contraintes géochronologiques applicables à l'évolution magmatique qui est associée au rift Midcontinental de 1,1 Ga. Les analyses d'une phase granophyrique de l'intrusion de Mineral Lake et du granite de Mellen, ces deux lithologies appartiennent au Complexe intrusif de Mellen, et d'une coulée de rhyolite étendue latéralement au sommet des Volcanites de Kallander Creek, ont procuré les âges $^{207}Pb/^{206}Pb$ moyens pondérés de 1102,0 ± 2,8 Ma (N = 2), 1100,9 ± 1,4 Ma (N = 5) et 1098,8 ± 1,9 Ma (N = 4), respectivement. Les analyses d'une coulée de rhyolite pyroclastique localisée au sommet des Volcanites de Porcupine ont fourni des âges $^{207}Pb/^{206}Pb$ qui varient de 1080 à 1137 Ma. Cette rhyolite montre qu'il y a une continuité entre les cristaux de zircon de forme complexe et ceux de forme prismatique beaucoup plus simple; ces derniers fournissent des résultats concordants et leur âge $^{207}Pb/^{206}Pb$ moyen pondéré est de 1093,6 ± 1,8 Ma (N = 2). Quatre cristaux prismatiques de zircon d'une rhyolite de la série des Volcanites de Chengwatana, dans le synclinal d'Ashland, ont donné des résultats disposés en une rangée linéaire qui intersecte la concordia à 1094,6 ± 2,1 Ma (indice des déviations pondérées = 1,3). Une autre rhyolite, que l'on croit appartenir à la série de Chengwatana, et qui a été récupérée dans les carottes du forage interceptant le horst de Hudson–Afton, dans le sud-est du Minnesota, a livré seulement une vingtaine de cristaux de zircon sans formes distinctives. Six analyses ont fourni des âges $^{207}Pb/^{206}Pb$ qui varient de 1112 à 1136 Ma, incluant une analyse d'une âge quasi concordant de 1130 Ma. Cependant, cet âge est considérablement plus ancien que celui obtenu pour les Volcanites de Chengwatana, dans le synclinal d'Ashland, et aussi de toutes les autres roches du rift Midcontinental qui ont été datées avec précision.
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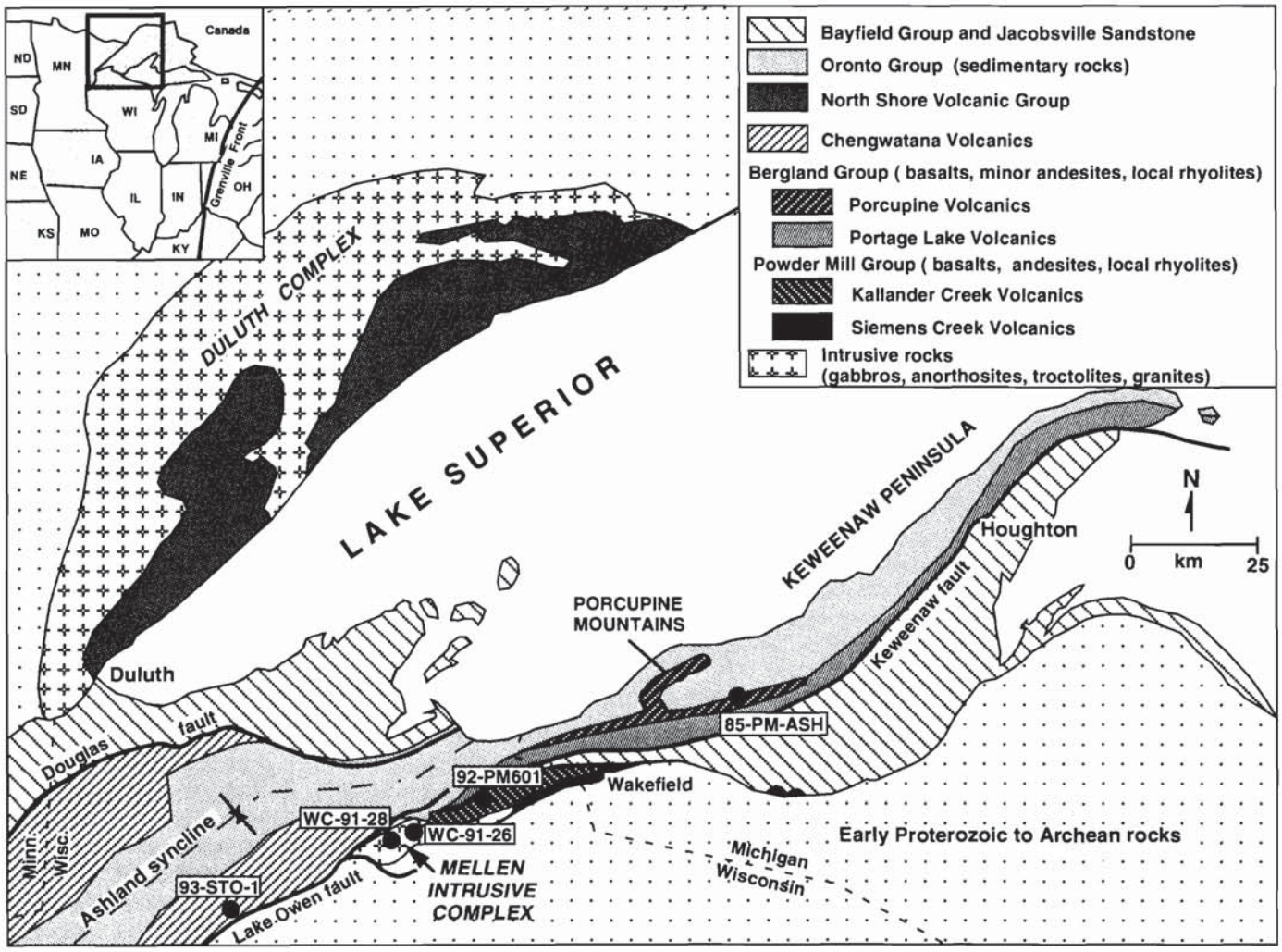
Introduction

The Keweenawan Supergroup volcanic and plutonic rocks of the 1.1 Ga Midcontinent rift system exposed south of Lake Superior have been the subject of extensive mapping (Butler and Burbank 1929; Cornwall 1954*a*, 1954*b*, 1954*c*, 1955; Cornwall and Wright 1954; Leighton 1954; Cornwall and White 1955; Olmsted 1968; Johnson and White 1969; Hubbard 1975*a*, 1975*b*; Klewin 1990), geophysical studies (Behrendt et al. 1988, 1990; Cannon et al. 1989; Chandler et al. 1989; Hinze and Braile 1990; Hutchinson et al. 1990; Cannon 1992), and geochemical studies (Cornwall 1951;

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Fig. 1. Generalized geology of western Lake Superior, showing location of samples, except for V-66-2, used for dating (also see Appendix). Stippled area in insert map is Midcontinent rift system.



Paces 1988; Paces and Bell 1989; Nicholson and Shirey 1990; Nicholson 1992; Seifert et al. 1992). Recent mapping by the United States Geological Survey of the area from Ontonagon, Michigan, west to Mellen, Wisconsin (Cannon et al. 1995, 1996) has refined the stratigraphy and structural relationships within the southern limb of the rift system, raising correlation and geochronologic questions that are amenable to investigation using high-precision U–Pb zircon dating techniques (Krogh 1982). In this study, we address the following questions: (1) What is the age of the Mellen Intrusive Complex and surrounding volcanic rocks? (2) What is the age of the Porcupine Volcanics and how does it constrain estimates of the duration of Portage Lake volcanism? (3) What is the age of the Chengwatana Volcanics and are they correlative with the Portage Lake Volcanics as postulated (Green 1982)?

In addition to two plutonic rocks from the Mellen Intrusive Complex, this study was carried out on rhyolites occupying key stratigraphic positions in the otherwise predominantly basaltic Keweenawan rocks. Previous geochronologic analyses of rhyolites elsewhere in the rift had revealed the presence of inherited older components in the zircon populations of some Keweenawan felsic rocks (Davis and Sutcliffe 1985; Palmer and Davis 1987). Using single-crystal techniques, older Proterozoic or Archean zircon components were indeed found in three of the four rhyolite samples, but by careful selection of single zircon grains based on crystal morphology and chemical parameters, it was possible to avoid composite grains. As in previous Keweenawan dating efforts (Paces and Miller 1993), the best test of a valid crystallization age is the ability to obtain several replicate, concordant results having $^{207}Pb/^{206}Pb$ ages lying within analytical uncertainty of each other.

Geologic setting

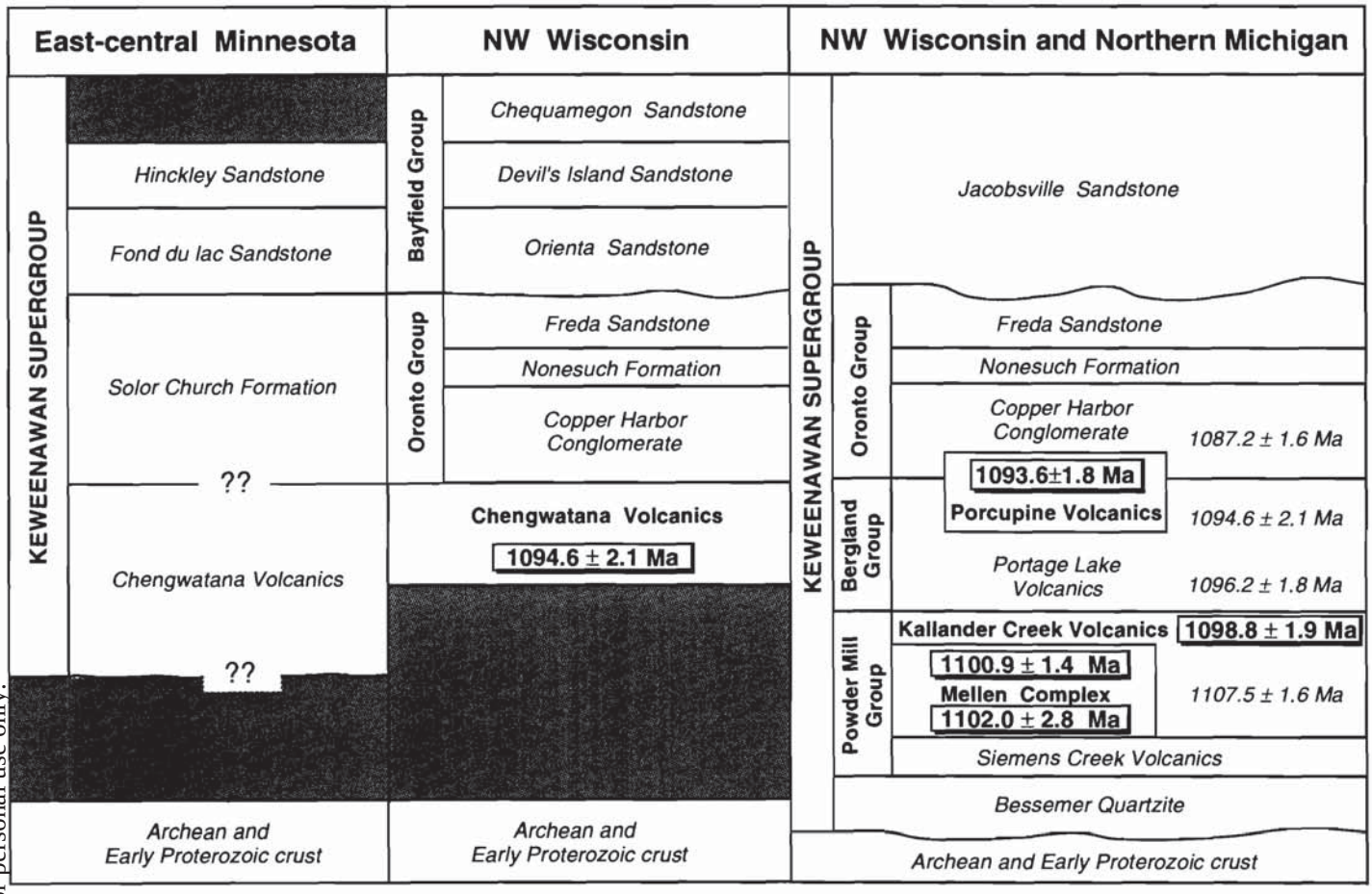
The igneous and sedimentary rocks of the Keweenawan Supergroup cropping out along the south shore of Lake Superior in northwestern Michigan and northwestern Wisconsin constitute the south limb of the Midcontinent rift (Figs. 1, 2). The rocks generally dip to the north at a shallow to moderate angle and are displaced southward along a number of thrust faults, which formed late in the rifting history when compressional forces caused transverse contraction of the basin. Despite some structural complications, a continuous stratigraphic column can be reconstructed for the rift-

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Fig. 2. Stratigraphic column of Keweenawan Supergroup rocks. Ages outlined in shadowed boxes are from this study; other ages for the Bergland and Oronto groups are from Davis and Paces (1990), and for the Powder Mill Group from Davis et al. (1995).



related rocks, which generally become progressively younger to the north.

The Middle Proterozoic Bessemer Quartzite (Fig. 2) forms the basal unit of the Keweenawan Supergroup. This quartz-rich fluvial and lacustrine sandstone was deposited in a shallow, regional basin and is generally considered to mark the initial stage of crustal sagging (Green 1982, 1983). Conformably overlying the Bessemer and representing the onset of volcanic activity are the tholeiitic pillow and flood basalts of the Siemens Creek Volcanics (~1.5 km thick) followed by the basalts, andesites, and rhyolites of the Kallander Creek Volcanics (2–4 km thick). These two extrusive units form the reversely magnetized Powder Mill Group, which crops out in a 70 km long belt in northwestern Wisconsin and adjacent Michigan (Books 1968; Hubbard 1975*b*; Palmer and Halls 1986). The upper contact of the Powder Mill Group is truncated to the west along the Lake Owen fault, and to the east it is in part cut out by the Keweenaw thrust fault and in part buried under the younger Jacobsville Sandstone (Fig. 1).

The Kallander Creek Volcanics reach their maximum thickness near Mellen, Wisconsin, and can be interpreted as having formed a shield volcano with considerable relief locally affecting the thickness and facies of subsequent overlapping strata (Cannon et al. 1993). A zircon U–Pb age of 1107.5 ± 1.6 Ma has been obtained on a pyroclastic rhyolite flow stratigraphically near the middle of the Kallander Creek Volcanics in western Michigan (Davis et al. 1995). The Mineral Lake intrusion and the Mellen granite, which together with the Potato River intrusion make up the Mellen Intrusive Complex (Leighton 1954; Olmsted 1968; Klewin 1990; Seifert et al. 1992), intrude the Kallander Creek Volcanics and are postulated to be approximately contemporaneous with the upper part of it (Cannon et al. 1993). The Mellen granite, which intruded and brecciated the more mafic units of the Mellen Intrusive Complex, cuts nearly all preserved parts of the adjacent Kallander Creek.

The Portage Lake Volcanics and all younger Midcontinent-rift-related rocks have normal magnetic polarity. The Portage Lake Volcanics (up to 5 km thick) extend parallel to the south shore of Lake Superior for 200 km from the vicinity of Mellen eastward to the tip of the Keweenaw Peninsula. The true thickness of the Portage Lake Volcanics is unknown, because its base is truncated by the Keweenaw fault. However, where exposed, the unit maintains a thickness of about 3 km, except where the formation pinches out to the west against the Kallander Creek Volcanics edifice and where it increases in thickness to about 5 km near the tip of Keweenaw Peninsula to the east. The Portage Lake Volcanics represent the most extensive exposure of the voluminous outpouring of high-Al tholeiite flood basalt associated with the Midcontinent rift, and for much of its strike length

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it is comprised of more than 200 individual basalt and minor intermediate and felsic flows. On Keweenaw Peninsula two particularly thick Portage Lake basalt flows with granophyric segregations, the Copper City flow near the Keweenaw fault and the Greenstone flow near the top of the formation, have yielded precise zircon U–Pb ages of 1096.2 ± 1.8 and 1094.0 ± 1.5 Ma, respectively (Davis and Paces 1990).

The Porcupine Volcanics (0–4 km thick) conformably overlie the Portage Lake Volcanics in the vicinity of the Porcupine Mountains in northwestern Michigan and adjacent Wisconsin. Although attaining considerable local thickness, this sequence of basalts, andesites, and rhyolites thins out rapidly along strike and is not mapped as a separate unit beyond 25 km east or west of the Porcupine Mountains. The physical morphology and geochemical characteristics of the Porcupine Volcanics suggest that they, like the Kallander Creek Volcanics, represent a local accumulation of flows (i.e., shield volcano). This differentiated sequence marks the end of basin-wide volcanism in the region, and the dating of Porcupine Volcanics was deemed important for establishing the duration of flood-basalt volcanism.

Conformably overlying these volcanic rocks is a predominantly sedimentary section, collectively named the Oronto Group, and recently reviewed by Daniels (1982). The lowest unit, the Copper Harbor Conglomerate (0–2 km thick), contains abundant clasts of basalt and rhyolite as well as various sedimentary rocks and, locally, thin basalt to andesite flows (informally referred to as the Lake Shore traps). One of these andesites from the tip of Keweenaw Peninsula has yielded a zircon U–Pb age of 1087.2 ± 1.6 Ma and represents the final stage of rift volcanism (Davis and Paces 1990). Gradational above the Copper Harbor Conglomerate is the Nonesuch Shale (up to 250 m thick), and, above it, the thick, arkosic Freda Sandstone (up to 3.5 km thick). On the basis of lithofacies interpretation, this sedimentary sequence reflects a tectonic and depositional transition from an extensional rift to a thermal subsidence basin environment.

To the west of the previously described section of Keweenawan rocks and, in part, separated from it by the Lake Owen fault is the northeast-plunging Ashland syncline, containing a lower volcanic sequence, the Chengwatana Volcanics, and an upper sedimentary sequence, the Oronto Group (Morey and Ojakangas 1982). Structurally, these rocks occupy the axial part of the Midcontinent rift south of the western Lake Superior region and, upon subsequent closure of the rift basin, were thrust southeastward against the more marginal facies rocks of the Powder Mill Group. Despite poor exposures and structural complexities, the Chengwatana Volcanics are believed to be correlative with the Portage Lake Volcanics to the east (Green 1982). The Oronto Group of the Ashland syncline can be traced into equivalent sedimentary rocks overlying the Porcupine and Portage Lake volcanics to the east. The Chengwatana Volcanics continue southward into the subsurface beneath Paleozoic platform cover where, presumably, they are encountered in several drill cores penetrating the Hudson–Afton horst in southeast Minnesota.

Analytical procedure

Rock sample size varied considerably, from 2 to 100 kg, based partially on expected zircon yield and partially on availability of collected material. Accordingly, the amount of recovered zircon ranged from a fraction of a milligram to over a gram, with the lowest yields coming from the two Chengwatana samples (93-STO-1 and V-66-2) and the greatest yield coming from the Kallander Creek Volcanics (92-PM-601). Only in the case of the Chengwatana core sample (V-66-2) was there an insufficient number of crystals available to adequately characterize the zircon population or to select morphologically distinct grains to evaluate the effects of inheritance. Generally, between 50 and 100 grains of zircon exhibiting the fewest fractures and inclusions were handpicked into a pneumatic abrader charged with pyrite (Krogh 1982). Abrasion continued for 15–20 h as the zircon was gradually reduced in size and number until only about a dozen well-rounded grains exceeding 200 mesh in size remained. Where morphological distinction existed within a sample, separate abrasion runs were made of the different grain types to insure their identification after external shape was lost. The approximately 20 grains of zircon recovered from sample V-66-2 were only lightly abraded and were not otherwise distinguished as to quality or shape.

The abraded zircon grains selected for analysis were leached in dilute HNO_3 to remove pyrite coating and other surface contamination acquired during the abrasion process. The individual grains, weighing between 1.8 and 24.7 μg, were transferred into 0.3 mL Teflon vials to which was added 10 μL 1 M HNO_3, 25 μL HF, and a 5–10 μL aliquot of mixed ^{205}Pb–^{235}U–^{230}Th spike. A puck containing seven sample vials was inserted into a high-pressure digestion bomb above a reservoir containing additional HF and placed into a 210°C oven, where the zircon dissolved in 3–5 days. The residue from each sample was dried, redissolved in 10 μL HNO_3 under pressure to insure equilibration between sample and spike, and again taken to dryness. No anion exchange chemistry was performed on these single-grain samples. Final preparation for mass spectrometry was accomplished by picking up the sample in 1–2 μL H_3PO_4 and several microlitres silica gel and loading the solution onto a rhenium filament. Total process blanks range between 1 and 3 pg Pb and between 0.1 and 0.2 pg for U and Th.

All isotopic data were obtained on a single-collector VG-54E thermal ionization mass spectrometer, and measurements were made using a Daly multiplier for ion detection. Because no chemical separation is performed prior to sample loading, U and Th are run as oxides from the same filament as the Pb. Appropriate corrections have been applied to the data for mass-dependent fractionation as well as Daly bias and nonlinearity. The U–Th–Pb data for 36 zircon analyses from six different rocks are presented in Table 1, and the best estimates of crystallization ages are summarized in Table 2. Results are also presented graphically on concordia diagrams in Fig. 3. All errors reported herein are quoted at the 95% confidence limit.

Raw $^{206}Pb/^{204}Pb$ ratios vary from 200 to 25 000 (over 800 for all but two analyses) and attest to the generally low chemistry blank levels and initial common Pb contents (Table 1). After subtracting a nominal Pb blank correction of 2 pg, over half of the analyzed samples are calculated to have 4 pg or less of residual common Pb, possibly attributable to ar underestimation of the blank or some minor impurity in the zircon. Six samples, all but one with U contents over

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Table 1. Zircon isotopic data for some Keweenawan Supergroup rocks from the south shore of Lake Superior.

Sample designation		Concentration (ppm)			Total common Pb (pg)[a]	$^{206}Pb/^{204}Pb$[a]	Isotopic composition of Pb (atom %)[b]				Age (Ma)[c]			
Grain No.	Mass (μg)	U	Th	Pb			^{204}Pb	^{206}Pb	^{207}Pb	^{208}Pb	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{208}Pb/^{232}Th$
WC-91-28: granophyric phase of Mineral Lake intrusion														
1	5.3	532.0	164.8	99.99	18.2	1 709	0.0437	83.21	6.958	9.791	1068±5	1079±4	1100±3	1100±12
2	2.4	330.5	185.2	64.78	3.8	2 247	0.0181	79.09	6.289	14.60	1064±5	1077±5	1102±4	1079±14
3	5.2	124.8	46.7	23.90	3.2	2 245	0.0157	83.27	6.572	10.14	1093±4	1096±4	1102±4	1084±24
4	10.1	93.5	26.0	17.22	2.9	3 522	0.0090	85.55	6.653	7.785	1083±5	1089±5	1102±4	1095±24
WC-91-26: Mellen granite														
1	6.7	200.3	—	40.76	4.7	3 108	0.0152	77.67	6.133	16.19	1083±5	1089±4	1100±4	—
2	5.9	177.4	108.3	37.45	10.6	1 125	0.0564	76.89	6.654	16.40	1101±6	1100±5	1099±5	1097±14
3	18.0	484.5	281.8	94.18	3.6	25 850	0.0015	79.48	6.067	14.46	1064±4	1075±4	1097±4	1066±12
4	7.0	281.6	189.3	57.32	6.8	3 224	0.0043	78.30	6.029	15.67	1095±5	1097±4	1101±4	1040±12
5	4.1	1356.1	848.8	270.3	6.5	9 236	0.0061	78.66	6.084	15.25	1078±5	1086±5	1102±3	1060±8
6	7.1	701.3	416.8	140.7	5.2	10 546	0.0048	78.79	6.073	15.13	1086±3	1091±3	1101±2	1116±14
92-PM-601: rhyolite at top of Kallander Creek Volcanics														
1	6.9	101.1	72.7	22.20	8.5	937	0.0624	74.62	6.566	18.75	1108±7	1105±6	1098±4	1110±22
2	10.0	94.2	72.3	20.23	5.5	1 947	0.0254	75.03	6.073	18.87	1100±5	1100±4	1099±5	1110±28
3	12.0	1448.3	1364.0	313.1	136.5	1 397	0.0505	71.60	6.156	22.19	1054±3	1067±2	1094±2	1034±6
4	4.8	42.4	40.2	9.47	2.7	847	0.0274	72.07	5.876	22.03	1098±9	1098±7	1099±5	1095±20
5	24.7	891.2	837.7	194.5	18.4	13 496	0.0049	73.07	5.634	21.29	1094±4	1096±3	1099±3	1084±18
85-PM-ASH: pyroclastic rhyolite at top of Porcupine Volcanics														
1	3.2	399.8	525.8	67.66	5.6	2 020	0.0246	74.49	6.085	19.40	876±6	949±5	1122±3	534±3
2	5.5	470.7	590.0	73.65	4.4	4 764	0.0090	74.56	5.916	19.51	818±5	909±5	1137±6	537±2
3	7.8	263.6	446.6	50.14	3.3	5 948	0.0053	72.12	5.592	22.29	952±3	1000±3	1108±3	556±4
5	11.2	206.3	196.2	43.87	7.7	3 286	0.0168	73.08	5.752	21.15	1066±18	1071±14	1080±10	1017±11
6	6.6	430.8	679.5	74.17	7.4	3 367	0.0160	72.33	5.781	21.88	868±6	941±5	1116±4	521±5
7	2.6	494.2	456.8	101.9	2.8	4 785	0.0053	73.52	5.648	20.83	1044±5	1059±4	1090±3	1020±7
8	6.5	225.9	184.5	47.99	4.7	3 502	0.0129	75.14	5.890	18.96	1094±3	1094±3	1094±2	1064±8
9	4.6	349.2	442.7	58.69	2.3	5 981	0.0026	74.94	5.754	19.30	880±4	946±4	1103±5	571±13
10	2.3	182.7	144.2	41.76	9.8	494	0.1179	71.45	7.101	21.33	1091±5	1092±5	1093±3	1089±8
93-STO-1: quartz-phyric rhyolite near base of Chengwatana Volcanics, Ashland syncline														
2	2.3	172.1	310.1	42.47	18.8	220	0.2475	59.89	8.093	31.77	954±6	1002±6	1107±7	694±6
3	7.4	1167.1	955.0	225.5	32.3	2 704	0.0258	74.08	5.961	19.98	986±8	1016±7	1083±5	996±7
4	4.0	427.8	326.4	90.27	8.8	2 142	0.0277	75.23	6.102	18.64	1085±3	1087±3	1092±4	1078±6
5	9.6	612.6	290.5	115.4	9.3	6 829	0.0095	81.61	6.316	12.07	1058±10	1068±8	1088±6	1031±11
6	3.4	320.1	549.1	63.97	10.6	971	0.0572	67.34	5.995	26.60	923±7	983±6	1120±7	638±4
7	10.7	1237.4	756.9	241.4	31.7	4 564	0.0163	79.25	6.245	14.49	1062±2	1072±2	1092±2	983±8
V-66-2: rhyolite from drill core in Chengwatana Volcanics below Solor Church Formation, Hudson–Afton horst														
1	7.7	324.0	341.3	56.38	5.0	4 718	0.0102	77.39	6.077	16.53	935±7	989±6	1112±4	597±6
2	6.4	304.5	203.8	53.76	4.1	4 582	0.0095	79.16	6.217	14.61	968±3	1015±3	1117±2	838±15
3	4.8	122.8	75.2	25.50	2.8	2 397	0.0114	79.39	6.301	14.30	1128±6	1129±5	1130±3	1043±5
4	3.3	148.2	89.9	28.44	2.4	2 114	0.0090	78.98	6.223	14.79	1044±6	1071±6	1126±5	1014±5
5	2.8	270.9	155.6	48.73	8.2	890	0.0673	77.01	6.857	16.06	949±8	999±7	1112±7	946±13
6	1.8	211.6	160.1	33.74	2.3	1 502	0.0062	79.21	6.233	14.55	882±8	958±8	1136±8	675±12

Notes: Decay constants: $^{238}U = 1.55125 \times 10^{-10}$ year^{-1}; $^{235}U = 9.8485 \times 10^{-10}$ year^{-1}; $^{232}Th = 4.9375 \times 10^{-11}$ year^{-1}; $^{238}U/^{235}U = 137.88$. Isotopic composition of 1.1 Ga common lead is assumed to be ^{204}Pb:^{206}Pb:^{207}Pb:^{208}Pb = 1:16.89:15.49:36.57 (Stacey and Kramers 1975).

[a]Measured; corrected for spike and fractionation only.

[b]Corrected for 2 pg of blank Pb.

[c]Ages corrected for blank and common Pb; errors are given at 95% confidence level.

500 ppm, have >10 pg common Pb, which likely resides in altered metamict domains or some exotic microinclusion. Initial Pb compositions are assigned the value of 1.1 Ga Stacey and Kramers (1975) model lead; however, some variation in this isotopic composition would result in only very minor effects on calculated ages due to the highly radio-

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Table 2. Summary of zircon U–Pb geochronological results.

Stratigraphic unit	Sample No.	Rock type	Weighted-average $^{207}Pb/^{206}Pb$ age (Ma)	Upper concordia intercept age (Ma)
Mellen Intrusive Complex				
Mineral Lake Intrusion	WC-91-28	Granophyre	1102.0±2.8 (N = 2)	1102.1±3.5 (MSWD = 0.9)
Mellen granite	WC-91-26	Granite	1100.9±1.4 (N = 5)	1101.5±2.9 (MSWD = 1.0)
Kallander Creek Volcanics	92-PM-601	Rhyolite	1098.8±1.9 (N = 4)	1099.0±2.6 (MSWD = 2.1)
Porcupine Volcanics	85-PM-ASH	Pyroclastic rhyolite	1093.6±1.8 (N = 2)	1093.8±1.4 (MSWD = 1.7)
Chengwatana Volcanics	93-STO-1	Quartz-phyric rhyolite	nd[a]	1094.6±2.1 (MSWD = 1.3)
Chengwatana Volcanics	V-66-2	Rhyolite	nd[a]	nd[b]

[a]Fewer than two analyses lie within 2% of concordance; thus, no weighted-average age was calculated.

[b]No upper concordia intercept age is reported due to very small sample size and a larger scatter in $^{207}Pb/^{206}Pb$ ages. The geological interpretation of a single concordant analysis at 1130 Ma is in question. See text for discussion.

genic nature of the Pb in these zircons. Weighted-average $^{207}Pb/^{206}Pb$ ages (weighted by the inverse variance of the individual ages) are calculated for those samples where two or more analyses lie within 2% of concordance. Following the precedent of Paces and Miller (1993), we accept these weighted-average $^{207}Pb/^{206}Pb$ ages as the best estimate of the time of host rock crystallization (Table 2). Ages have also been calculated using the PBDAT and ISOPLOT programs of Ludwig (1991*a*, 1991*b*) whereby upper intercepts with concordia are calculated by a linear regression of the error-weighted individual analyses. In all cases where linear regression ages are calculated, the resulting mean square of the weighted deviates (MSWD) has values between 0.9 and 2.1, suggesting that, at most, only minor scatter exists in the data outside of analytical errors.

Although $^{208}Pb/^{232}Th$ ages are also calculated and presented in Table 1, these ages are not individually discussed because they often exhibit greater analytical error and are intrinsically more ambiguous in evaluating discordance. Suffice it to note the general agreement of essentially concordant U–Pb ages with their corresponding Th–Pb ages, whereas increasingly discordant U–Pb ages are usually accompanied by an even greater Th–Pb age disturbance. Interestingly, many of the highly disturbed zircons also display unusually low U/Th ratios (>1.0), which may have petrogenetic significance or, alternatively, simply reflect the relative elemental mobility (Pb > U > Th) associated with the alteration process causing the age discordance. An apparent correlation between U/Th ratio and presence of an inherited component in the zircon could likewise be explained as either an original geochemical or superimposed disturbance feature. Because the focus of this paper is on the geochronological results derived from the more concordant analyses, we do not pursue these interesting possibilities further.

Results

Mellen Intrusive Complex

To test the hypothesis of contemporaneity among the intrusive and extrusive rocks in the Mellen area, samples of granophyre from the Mineral Lake intrusion, granite from the Mellen granite, and the extrusive rhyolite from the top of Kallander Creek Volcanics were collected for dating. The morphologically least complicated zircon was obtained from a granophyric phase of the Mineral Lake intrusion and from the Mellen granite. Each of these rocks appears to contain a single population of colorless to pale yellow, euhedral zircon consisting mostly of simple prismatic and dipyramidal forms. No effort was made to differentiate the zircon from these two plutonic rocks into subgroups based on shape or size, unlike the rhyolites whose zircon displayed a more complex morphology and crystallization history. Approximately 50 grains of the clearest, least fractured and included, unbroken crystals from the −50+150 mesh fractions were abraded, which resulted in concordant to only slightly discordant analyses (Table 1). The narrow range in U/Th displayed by the individual zircon grains from the Mellen granite and the Kallander Creek Volcanics also suggests a close chemical affinity within each population. By contrast, a much greater variation of this ratio encountered in the Mineral Lake intrusion may reflect a rapidly changing U/Th in residual granophyric melt. The abrasion process did not discriminate particularly well according to uranium content, judging by the large variation (up to an order of magnitude) shown in this element among the analyzed grains.

The interpretation that these plutonic rocks hosted a simple, homogeneous zircon population is supported by the close agreement of $^{207}Pb/^{206}Pb$ ages for all analyses (Table 1). Including only analyses that lie within 2% of concordancy, the weighted average of the $^{207}Pb/^{206}Pb$ ages for two (grains 3 and 4) out of four analyses from the Mineral Lake intrusion is 1102.0 ± 2.8 Ma and five (grains 1, 2, 4, 5, and 6) out of six analyses from the Mellen granite is 1100.9 ± 1.4 Ma. By comparison, a linear regression through all of the data for the Mineral Lake intrusion and the Mellen granite give concordia upper intercept ages of 1102.1 ± 3.5 Ma (MSWD = 0.9) (Fig. 3A) and 1101.5 ± 2.9 Ma (MSWD = 1.0) (Fig. 3B), respectively.

Kallander Creek Volcanics

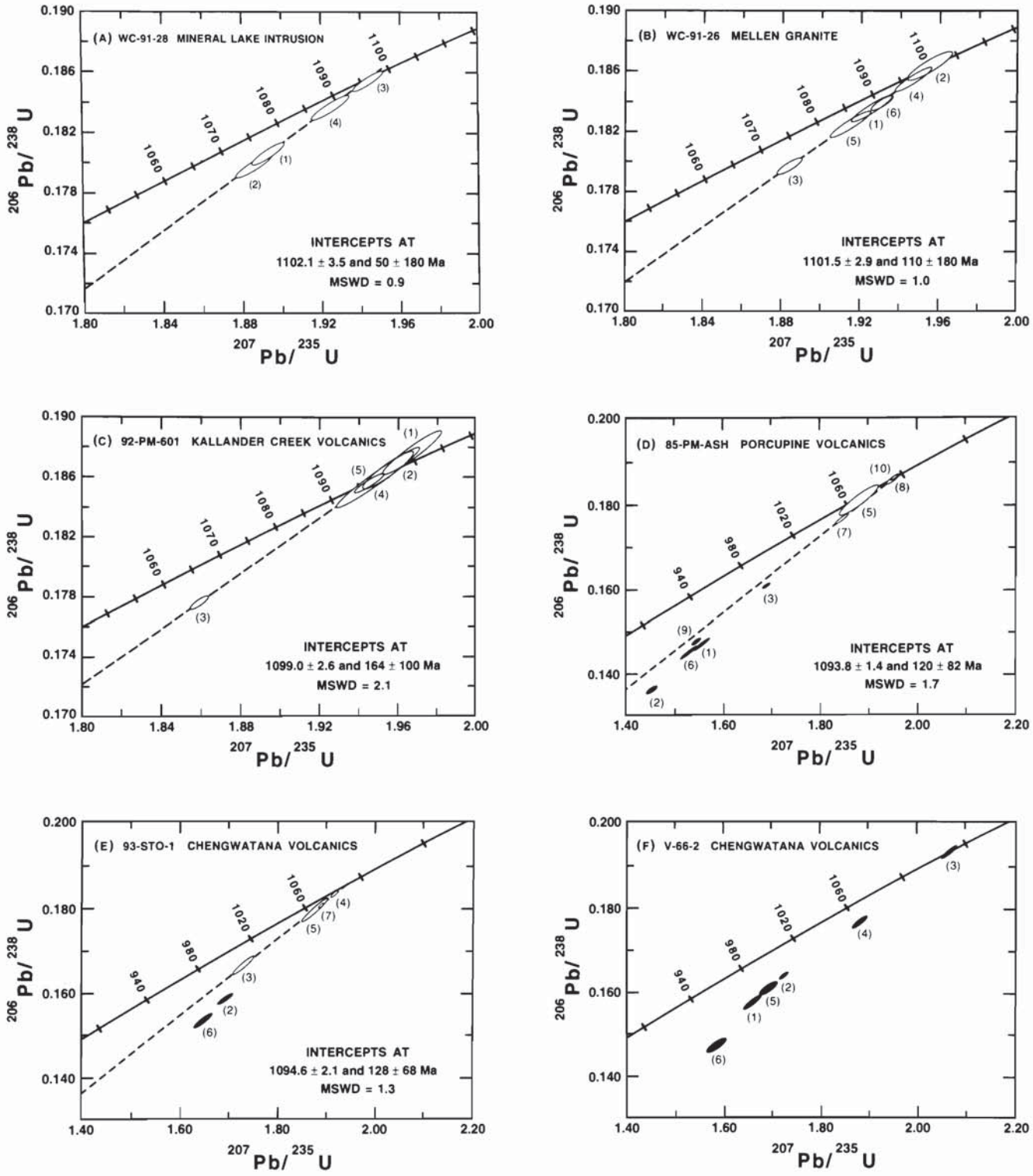
Our initial experience at dating zircon from Keweenawan rhyolites proved to be deceptively simple compared with subsequent efforts. Although morphologically somewhat more complex than those from the presumably related plutonic rocks, the analyzed zircon grains from the Kallander Creek Volcanics, also colorless to pale yellow, were chosen entirely on the basis of internal clarity and freedom from defects (Fig. 4). As with the plutonic zircon, approximately 50 grains of the best crystals from the −50+150 mesh frac-

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Fig. 3. Concordia diagrams for Keweenawan Supergroup rocks. Numbers given for error ellipses correspond to zircon grain numbers listed in Table 1. Error ellipses and regression age uncertainties represent 95% confidence limits. Note that for samples WC-91-26, WC-91-28, 92-PM-601, and 85-PM-ASH, two or more analyses lie within 2% of concordance. Thus, for these four samples, a weighted-average $^{207}Pb/^{206}Pb$ age was calculated (Table 2) and used in the geological interpretation.



tion were abraded with the analyses made on the final surviving grains. Again, close agreement of $^{207}Pb/^{206}Pb$ ages suggests a straightforward crystallization age, despite a variation of more than a factor of 30 in uranium content. Here, four (grains 1, 2, 4, and 5) out of five of the analyses are concordant to within 1% and give a weighted average $^{207}Pb/^{206}Pb$ age of 1098.8 ± 1.9 Ma. A linear regression through all of the data gives a concordia upper intercept age of 1099.0 ± 2.6 Ma (MSWD = 2.1) (Fig. 3C). Thus, the data are compatible with the interpretation of Cannon et al. (1993) that the units dated in this study constitute a genetically related volcanic–plutonic complex.

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Fig. 4. Photomicrograph illustrating morphology of −150+250 mesh fraction zircon from sample 92-PM-601, rhyolite at top of Kallander Creek Volcanics. Internally, the grains range from almost clear to highly fractured, turbid, and sometimes zoned. Note uneven development of dipyramidal faces on some of the crystals.

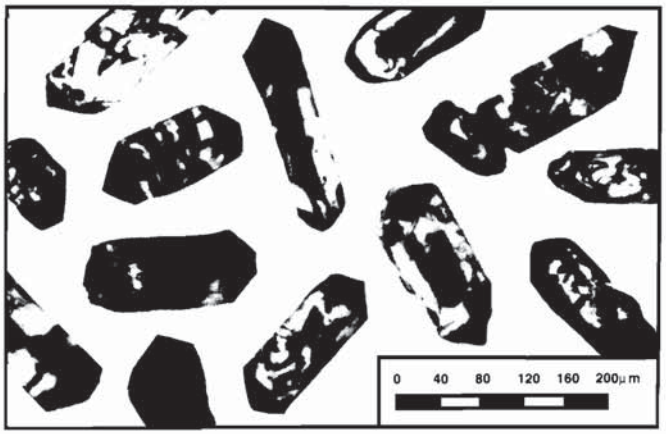
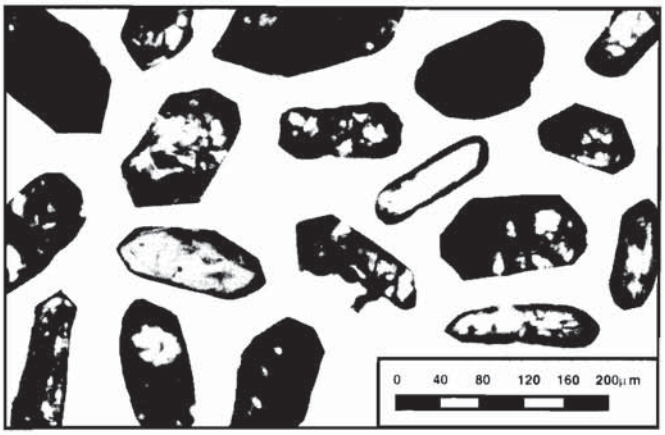


Fig. 5. Photomicrograph illustrating morphology of −150+250 mesh fraction zircon from sample 85-PM-ASH, pyroclastic rhyolite at top of Porcupine Volcanics. Although the internal and external features of the grains are similar to those of sample 92-PM-601, some of the darker, more complex crystals contain an older, inherited zircon component.



Porcupine Volcanics

The first indication of anomalously old $^{207}Pb/^{206}Pb$ ages attributable to an inherited zircon component arose from a pyroclastic rhyolite flow near the top of the Porcupine Volcanics. Unlike the simple behavior of the Kallander Creek zircon, the initial five analyses from a small but ample suite of Porcupine Volcanics zircon (grains 1, 2, 3, 5, and 6) showed a wide range in $^{207}Pb/^{206}Pb$ ages (1080−1137 Ma) that, with the exception of grain 5, were too old for the stratigraphic position of this unit. These analyses were determined on +150 mesh zircon that had been selected uncritically with regard to external morphology, prompting us to examine the total zircon population in more detail in search of clues to the apparently aberrant initially observed behavior. Although no discrete bimodality in shape was observed, a continuum can be recognized from morphologically complex crystals prominent throughout the population to prismatic, generally more elongate crystals present mainly in the smaller size fractions (Fig. 5). The most complex crystals commonly displayed several dipyramidal forms of uneven development with or without any obvious prism faces. A color distinction also exists, with the complex crystals dominated by dark yellowish-brown grains and the prismatic crystals by colorless to pale yellow grains.

Subsequent analyses concentrated on the prismatic crystals and confirmed their identity as a distinct morphological variety. Contrary to the initial analyses, which are presumed to have been performed on the larger and more complex grains, three out of four of the prismatic grains yield tightly grouped $^{207}Pb/^{206}Pb$ ages. Two (grains 8 and 10) out of the four analyses having $^{207}Pb/^{206}Pb$ ages less than 1095 Ma are concordant to within analytical uncertainty and give a weighted average $^{207}Pb/^{206}Pb$ age of 1093.6 ± 1.8 Ma. A linear regression through the data for the four analyses (grains 5, 7, 8, and 10) gives a concordia upper intercept age of 1093.8 ± 1.4 Ma (MSWD = 1.7) (Fig. 3D). A perfect dichotomy in isotopic behavior between the two morphological types probably does not exist, however, in that grain 5 was not originally recognized as a prismatic crystal, and grain 9, chosen to be a prismatic crystal, gives a slightly anomalously old $^{207}Pb/^{206}Pb$ age.

Chengwatana Volcanics

Two samples of rhyolite within the Chengwatana Volcanics were selected to test the correlation between this thick volcanic sequence occupying the Ashland syncline and the Portage Lake Volcanics across the Lake Owen fault. One sample of quartz-phyric rhyolite was collected from near the base of the Chengwatana section on the southeast limb of the syncline near (<5 km) the Lake Owen fault. The other sample was obtained from a section of drill core that intersected Keweenawan rocks in the Hudson−Afton horst southeast of Minneapolis, Minnesota. The Hudson−Afton horst is a small block of volcanic rock uplifted during Cambro-Ordovician time that defines the southeast side of the much larger St. Croix horst, a rift-related feature that underlies much of southeastern Minnesota. The volcanic rocks from this core have been assigned to the Chengwatana Volcanic Group, and are overlain in adjacent drill holes by sandstone of the Solor Church Formation, a unit that is broadly correlated with the Oronto Group in the Ashland syncline.

The zircon analyses for the two samples of Chengwatana Volcanics also display complex isotopic systematics. The quartz-phyric rhyolite (sample 93-STO-1) from the lower part of the Chengwatana Volcanics exposed in the Ashland syncline provided +150 mesh zircon with a $^{207}Pb/^{206}Pb$ age span of 1083−1120 Ma. This broad age spectrum containing excessively old $^{207}Pb/^{206}Pb$ ages closely resembles that found for the Porcupine Volcanics rhyolite (sample 85-PM-ASH). A similar continuum of zircon morphologies is present, with both complex, dark yellowish-brown crystals and prismatic, colorless to pale yellow crystals present throughout the population (Fig. 6). Particularly noteworthy, the prismatic zircon varies widely in elongation, with length-to-width ratios ranging from about 3 to greater than 10. As was the case with the zircon from the Porcupine Volcanics rhyolite, the old $^{207}Pb/^{206}Pb$ ages appeared to be most closely associated with the morphologically complex crys-

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tals, and, accordingly, prismatic crystals were selected for later analyses. Two of the prismatic crystals (grains 4 and 7) were chosen because of their extreme elongation (length-to-width ratio greater than 5) and produced the two most concordant analyses. Restricting a linear regression to data for analyses (grains 3, 4, 5, and 7: from 1 to 10% discordant) having $^{207}Pb/^{206}Pb$ ages less than 1100 Ma, a concordia upper intercept age of 1094.6 ± 2.1 Ma (MSWD = 1.3) (Fig. 3E) was calculated and is used as the preferred age of the unit.

Only about 20 pale to medium yellow, euhedral to subhedral zircon grains were recovered from sample V-66-2, a rhyolite encountered in drill core just below the Solor Church Formation. No clear morphological distinctions could be made among the limited number of zircon grains, which generally resembled those forming the middle of the continuum of the other rhyolites, nor was it possible to acquire a tight grouping of $^{207}Pb/^{206}Pb$ ages. All six analyses determined on this sample gave $^{207}Pb/^{206}Pb$ ages in excess of 1100 Ma, ranging from 1112 to 1136 Ma (Fig. 3F). One analysis (grain 3) does yield a virtually concordant age of 1130 Ma, which ordinarily might be considered excellent evidence for the crystallization age of the zircon and, presumably, the host rhyolite. However, this age is much older than that obtained for the Chengwatana Formation in the Ashland syncline, and seems to suggest that here, too, inherited zircon is present in the rhyolite.

Discussion

Age of the Mellen Intrusive Complex and adjacent rocks

The Kallander Creek Volcanics can be divided into two informal members, designated as upper and lower Kallander Creek Volcanics, based on differences in chemical characteristics, volume of intermediate rocks, and eruptive style. The lower member is dominated by basalts and basaltic andesites with minor andesite and rhyolite; flows typically were erupted as flood basalts. The upper member is dominated by intermediate rocks and includes several laterally extensive rhyolite flows. The only previously determined radiometric age for the Kallander Creek Volcanics was a U–Pb zircon age of 1107.5 ± 1.6 Ma obtained on a rhyolite tuff near the top (2.5 km above the base) of the lower member of the formation (Davis et al. 1995). The newly determined age for the rhyolite considered to be the top of the upper member is 1098.8 ± 1.9 Ma.

The upper member of the formation is believed to represent the volcanic edifice related to emplacement of the Mellen Intrusive Complex (Cannon et al. 1993). New ages of 1102.0 ± 2.8 Ma for a granophyre from the Mineral Lake intrusion and 1100.9 ± 1.4 Ma for the Mellen granite are indistinguishable, thus implying that the two intrusive rocks are essentially coeval, although not necessarily comagmatic. The upper Kallander Creek rhyolite also is indistinguishable in age from the intrusive rocks, thus supporting the possibility that the upper portion of the Kallander Creek Volcanics indeed may be related to the intrusive rocks. Although this rhyolite could be 1–2 Ma younger than the Mellen granite, additional dating will be required to resolve the precise temporal relationship between flow and intrusion.

Dating of the uppermost Kallander Creek rhyolite also provides constraints on the timing of the shift in magnetic polarity from reversed to normal that has been recognized throughout the Lake Superior region. On the south shore, the polarity shift from reversed to normal historically has been placed at the boundary between the Kallander Creek Volcanics and the overlying Portage Lake Volcanics (e.g., Books 1968). However, a recent reconnaissance magnetic polarity survey identified the shift from reversed to normal magnetic polarity as occurring a few flows below what is considered to be the uppermost rhyolite in the Kallander Creek Volcanics. Thus, the newly determined age for this rhyolite constrains the minimum age of the regionally documented reversal to being no younger than 1098.8 Ma. This corroborates dates from the Duluth Complex that provide a lower limit of 1099 Ma on the age of the magnetic reversal (Paces and Miller 1993) and from the North Shore Volcanic Group that suggest the reversal is no younger than 1100.5 Ma (Davis et al. 1995).

The new age date for the top of the Kallander Creek Volcanics leads to the implication that the Kallander Creek (at most 4 km thick) was erupted over a period of 9–10 Ma (from about 1108 to 1099 Ma). Because the region into which the Mellen Intrusive Complex was emplaced apparently did not subside much after eruption of the lower Kallander Creek Volcanics and thus subsequently acted as topographic high, it has been postulated that one or more substantial breaks in basaltic volcanism (Cannon et al. 1993; Miller et al. 1995) may occur between the two dated Kallander Creek horizons. In addition, based on ages for flows in the North Shore Volcanic Group, Miller et al. (1995) and Davis et al. (1995) have postulated a hiatus of at least 5 Ma in basaltic eruptions during this time period. However, about half of the 18 km of basalts buried beneath western Lake Superior are magnetically reversed (Allen 1994), suggesting that volcanism did not cease altogether along the rift axis during this time. Likewise, Mariano and Hinze (1994) postulate that about 80% of the 15 km of

Fig. 6. Photomicrograph illustrating morphology of −150+325 mesh fraction zircon from sample 93-STO-1, quartz-phyric rhyolite near base of Chengwatana Volcanics, Ashland syncline. Note, again, general similarity in internal and external features of the grains to those of sample 92-PM-601 and 85-PM-ASH. As with sample 85-PM-ASH, an older, inherited zircon component is present in some of the darker, more complex crystals.



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basalts underlying eastern Lake Superior are reversed, thus implying substantial volcanic activity prior to 1100 Ma in that part of the rift. The postulated volcanic hiatus recorded in the Kallander Creek Volcanics may reflect the effect of a local region of nondeposition of basalt flows due to a topographic high in the vicinity of the Mellen Intrusive Complex rather than an actual break in flood basalt production within the central graben of the rift.

Duration of Portage Lake volcanism

The Portage Lake Volcanics represent the voluminous eruption of olivine tholeiitic basalt during the later stages of active rift extension (Paces 1988). Davis and Paces (1990) determined zircon U–Pb ages of 1096.2 ± 1.8 and 1094.0 ± 1.5 Ma for two flows near the bottom and top, respectively, of the exposed 5 km thick section of Portage Lake Volcanics on Keweenaw Peninsula. In this area the base of the formation is truncated by the Keweenaw fault. West of Keweenaw Peninsula, the Keweenaw fault dies out and the uppermost Kallander Creek rhyolite dated at about 1099 Ma provides a maximum age for the Portage Lake Volcanics. However, the Mellen Intrusive Complex and Kallander Creek shield volcano formed a topographic high at the time of deposition of the Portage Lake Volcanics, which thin markedly as the flows lap up against the edifice (Cannon et al. 1996). Probably, only the youngest flows of the Portage Lake Volcanics are preserved close to the Mellen Intrusive Complex, and there may be a several million year period of minimal extrusion or even an erosional hiatus between the Kallander Creek Volcanics and the lower part of the local Portage Lake Volcanics (Cannon et al. 1993).

In the Porcupine Mountains area, the Portage Lake Volcanics are overlain by the Porcupine Volcanics, comprising a local shield volcano edifice similar to that of the upper Kallander Creek Volcanics. The Porcupine Volcanics underlie the Copper Harbor Conglomerate, and represent the last significant volcanic activity in this area. Thus, the age determined for a rhyolite at the top of the Porcupine Volcanics (1093.6 ± 1.8 Ma) provides an upper limit both for the construction of the shield volcano and for the end of flood-basalt volcanism. This finding is consistent with previously published dates for the top of the underlying Portage Lake Volcanics.

Age of the Chengwatana Volcanics

The great thickness of Portage Lake flood basalts present in western Lake Superior region pinch out entirely westward against the Mellen Intrusive Complex and Kallander Creek shield volcano. At the present erosion level, this topographic high served to isolate the Portage Lake Volcanics on the east from the Chengwatana Volcanics on the west. As a consequence, the Portage Lake Volcanics cannot be traced directly into the Ashland syncline (Fig. 1), although the Chengwatana Volcanics are postulated to be equivalent to the Portage Lake Volcanics by virtue of their stratigraphic position underlying the Oronto Group sedimentary rocks. Current studies of major- and trace-element compositions of the Portage Lake Volcanics and Chengwatana Volcanics indicate significant chemical similarities between the two units. This correlation is also now corroborated by the age determination of 1094.6 ± 2.1 Ma for a rhyolite near the base of the Chengwatana Volcanics in the Ashland syncline.

The age of a subsurface rhyolite from the Hudson–Afton horst in southeastern Minnesota, which is usually considered an extension of the Chengwatana Volcanics of the Ashland syncline, is left in considerable doubt by the ambiguous geochronological results obtained on a core sample. The penetrated Keweenawan rocks reveal a volcanic section comprised predominantly of basalts with minor rhyolite, which is overlain, in ascending order, by the Solor Church Formation, Fond du Lac Sandstone, and Hinckley Sandstone. The sedimentary rocks appear to correlate with the Oronto and Bayfield groups (Fig. 2). However, the anomalously old and considerably scattered $^{207}Pb/^{206}Pb$ ages obtained on the rhyolite core sample fail to confirm a temporal equivalency with the newly determined age of the Chengwatana of the Ashland syncline. It is entirely possible that the inherited zircon component encountered in the rhyolites to the north is also present here, and in fact so dominates the sampled population that none of the analyzed grains avoided it.

Presumably, the discordant zircon analyses from all of the rhyolites in this study yielding $^{207}Pb/^{206}Pb$ ages greater than 1100 Ma identify composite grains comprised of an inherited older core and an overgrowth of volcanogenic zircon. If the core component derives from Early Proterozoic or Archean rocks underlying the Midcontinent rift system, the minor increase in $^{207}Pb/^{206}Pb$ ages suggests either an almost complete resetting of the age of the inherited component or a small core-to-rim ratio. Furthermore, the distinctive low U/Th ratio referred to previously may mirror differences in the chemistry of the evolving rhyolite magma. For example, low U/Th ratios could identify a lower crustal granulitic source whence the inherited core originally crystallized, whereas higher U/Th ratios could identify later growth in the entraining magma. Alternatively, without better characterization of the zircon, we must also entertain the possibility that the inherited component is derived from underlying Keweenawan rocks of only slightly older age ($\geq$1130 Ma). Little is known about older Keweenawan rocks buried as much as 20 km in deeper parts of the rift basin, and some or all of the anomalous zircon conceivably might be xenocrysts derived from such underlying volcanic or plutonic rock. Even more speculative would be the acceptance of the one essentially concordant analysis of about 1130 Ma as supportive of such an age for the subsurface rhyolite sample itself. In that case, the stratigraphy encountered within the drill core will certainly require major revision.

Conclusions

(1) The age of the Mellen Intrusive Complex is about 1101–1102 Ma, considerably younger than the age of about 1107 Ma determined for the middle of the Kallander Creek Volcanics, the rocks that the Mellen Intrusive Complex intrudes. However, the ages of the intrusive rocks are nearly indistinguishable from the age of 1099 Ma determined for the uppermost flow in the Kallander Creek Volcanics, thus allowing that the upper portion of the Kallander Creek Volcanics may be related to the intrusive rocks.

(2) The uppermost Kallander Creek rhyolite displays normal magnetic polarization, providing a minimum age of about

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1099 Ma for the regional magnetic polarity shift from reverse to normal, corroborating ages for the polarity shift determined for the North Shore Volcanic Group.

(3) An ash-flow rhyolite at the top of the Porcupine Volcanics yields an age of about 1093 Ma. Because the Porcupine Volcanics overlie the flood basalts of the Portage Lake Volcanics, this age constrains the cessation of widespread flood-basalt volcanism on the south shore of Lake Superior.

(4) An age determined for a rhyolite within the Chengwatana Volcanics in the Ashland syncline is about 1094 Ma, suggesting that at least part of the Chengwatana Volcanics is time equivalent to the Portage Lake Volcanics.

(5) An age determined for a rhyolite recovered from drill core within the Hudson–Afton horst along the eastern margin of the southward, subsurface projection of the Ashland syncline is apparently anomalously old (1130 Ma) with respect to the assignment of the rhyolite to the Chengwatana Volcanics. The reason for this older age has not been resolved but several possible causes can be entertained. (*i*) The dated sample comprised composite grains, including a component from older rocks, and the age reflects this inherited component. (*ii*) The zircon is xenocrystic and has been obtained from a slightly older volcanic or plutonic unit underlying the outcropping Keweenawan Supergroup. (*iii*) The rhyolite has been correctly dated, but it is not coeval with the Chengwatana Volcanics and implies an earlier onset of rift-related volcanism.

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Appendix: Sample locations and descriptions

91-WC-28

Granophyric phase of Mineral Lake intrusion (Ashland County, Wis.). Location: lat. 46°22′20″N, long. 90°52′00″W, SW 1/4, NW 1/4, sec. 34, T45N, R4W. Description: Roadside outcrop. Fine- to medium-grained granophyre comprised of quartz, microcline, hornblende, and biotite.

91-WC-26

Mellen granite (Ashland County, Wis.). Location: lat. 46°22′30″N, long. 90°44′30″W, SE 1/4, NW 1/4, sec. 21, T45N, R3W. Description: Roadside outcrop. Medium-grained granite comprised of quartz, microcline, plagioclase, biotite, and oxides, with coarser feldspar phenocrysts.

92-PM-601

Rhyolite at top of Kallander Creek Volcanics (Iron County, Wis.). Location: lat. 46°30′00″N; long. 90°22′30″W, NE 1/4, NE 1/4, sec. 4, T46N, R1E. Description: Outcrop along pipeline. Maroon rhyolite with small quartz and feldspar phenocrysts. Flow-banded rhyolite extends from Copper Falls, Wis., east about 30 km and is about 300 m thick. Basal few metres contain rhyolite blocks up to 20 cm across.

85-PM-ASH

Rhyolite from top of Porcupine Volcanics (Ontonagon County, Mich.). Location: lat. 46°43′N, long. 88°26′, NE 1/4, SW 1/4, sec. 21, T50N, R41W. Description: Quarry exposure. Lavender to dark brown rhyolite with numerous inclusions and fiamme, small

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quartz and feldspar phenocrysts, strongly flow banded. Laterally extensive ash flow overlies basalts and andesites and underlies Copper Harbor Conglomerate. Quarry appears to have exposed the rhyolite near its vent area as demonstrated by convoluted folding of flow units.

93-STO-1

Rhyolite from near base of Chengwatana Volcanics (Sawyer County, Wis.). Location: lat. 46°05′45″N, long. 91°30′05″W, NE 1/4, SW 1/4, sec. 28, T42N, R9W. Description: Outcrop along shore of Nelson Lake. Reddish-brown aphyric rhyolite with very sparse and minute quartz and feldspar phenocrysts. Some flow banding present. No contacts exposed.

V-66-2

Rhyolite from drill core in Chengwatana Volcanics (Dakota County, Minn.). Location: lat. 44°41′30″N, long. 93°01′00″W, SW 1/4, SW 1/4, sec. 8, T114N, R8W. Description: Core from Vermillion 66-2 well drilled by Northern Natural Gas Company in 1966. Mount Simon Sandstone overlies rhyolite with an intervening 1.6 m weathered zone. Hole bottomed in rhyolite. Seven rhyolite flows were recognized in 55 m of penetrated rhyolite; sample from flow 5. Dark reddish-brown aphyric rhyolite with sparse minute feldspar phenocrysts.

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